Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Physicist (PhD) with over ten years of experience building the simulations, datasets, and analysis code that test scientific ideas against data. Currently an expert evaluator of frontier language models: writing hard physics problems that stump them and diagnosing where their reasoning fails. Background in physics-based numerical modeling (PDE solvers, radiative transfer, MHD), scientific machine learning, and open-source research software adopted by other groups. Postdoctoral experience leading teams and mentoring students, and a habit of explaining technical results clearly to mixed audiences.

Skills

Languages: Python, C/C++, Fortran 95, Rust, R, Java, SQL, Julia, Shell
ML & Data: Scientific machine learning, hybrid physics–ML modeling, PyTorch, TensorFlow, JAX (introductory coursework), scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, neural networks, random forest, SVM
Modeling: Numerical PDEs (advection–diffusion, Fokker–Planck), finite difference/volume methods, multiphysics simulation, uncertainty quantification
HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5
AI Evaluation: Expert-level problem and benchmark design, prompt writing, failure-mode analysis of LLM reasoning, structured feedback, AI-assisted software development with test-based validation
Cloud & Tools: Git and code review, Docker, Azure, CI/CD (Jenkins), Kafka, Jira, QGIS, ParaView, VisIt

Experience

Independent Contractor, *Mercor* Jan 2026 – Present
Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.
- Diagnosed failure modes in model reasoning and wrote structured feedback to improve scientific accuracy.

Spatial Data Scientist, *TealWaters* May 2025 – Jan 2026
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km^2 Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Independent Researcher & Open-Source Developer, *Self-directed* Jan 2024 – May 2025
Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, an open-source Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer for relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library for wind-driven spore dispersal over hilly forest terrain (RcppParallel, OpenMP), using AI coding assistants (Claude), validated by a unit-test suite.

Software Engineer, *Paychex* Apr 2022 – Jan 2024
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Postdoctoral Research Associate, *Rochester Institute of Technology* Feb 2021 – Apr 2022
HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team in an NSF-sponsored upgrade of PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries, adding black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, identifying performance bottlenecks in the parallel code to make it run efficiently on HPC clusters.
- Calibrated Paramo parameters against Fermi-LAT observations using gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow, *Purdue University* Oct 2018 – Nov 2020
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer. Optimized with OpenMP; GPU offload tested with OpenACC.

- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; released and documented Paramo as open source, and other groups have adopted it, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals, one funded ($68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo* Jan 2018 – Sep 2018

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Graduate Research Assistant, *Universitat de València* Oct 2011 – Jul 2017

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Projects

Blazar Population ML Jan 2026 – Present

Physics-informed ML (Python, scikit-learn, PyTorch) github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Building a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, to be validated against Tleco output (in progress).
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

Selected Publications (9 peer-reviewed articles, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

Education

Universitat de València, *Ph.D. in Physics, Excellent Cum Laude* Jul 2017

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics* Sep 2011

Universidad Autónoma del Estado de México, *B.Sc. in Physics* Dec 2008